

Task Title: AI Task Ingestion Service & TANTRA Runtime Convergence Certification

Department: Web Development

Assignee Candidate: Rudra Parmeshwar

Priority: High

Target Date: 2026-08-30

Overview & Description

Develop and certify the AI Task Ingestion Service and validate that all independently developed BHIV services converge into a single deterministic TANTRA runtime. The task engine must process user document uploads (PDF, DOCX, TXT, MD), execute automated metadata extraction for candidate Rudra Parmeshwar, route candidate tasks within the Web Development department, and certify service orchestration, runtime contracts, constitutional boundaries, observability, replayability, and production readiness.

Mission

Certify that all independently developed BHIV services converge into a single deterministic TANTRA runtime by validating service orchestration, AI task ingestion (PDF, DOCX, TXT, MD document parsing), runtime contracts, constitutional boundaries, observability, replayability, and production readiness.

Rudra does not build business features. He certifies that every constitutional owner has integrated correctly.

Phase 1 – Runtime Service Convergence

Validate orchestration between:

- * NIYANTRAN
- * PARIKSHAK
- * MasterDB
- * MDU
- * Bucket
- * InsightFlow
- * TANTRA Runtime

Ensure every service communicates only through approved APIs and runtime contracts.

Phase 2 – Runtime Contract Certification

Validate:

- * Multi-Format Ingestion (PDF, DOCX, TXT, MD document parsing)
- * Automated Metadata Extraction & Candidate Routing
- * API compatibility
- * Authentication
- * Trace IDs (x-trace-id)
- * Version compatibility
- * Error propagation
- * Retry behaviour
- * Deterministic execution & SHA-256 hashing

No service may bypass another service's published contract.

Phase 3 – Constitutional Ownership Validation

Verify:

- * No duplicated task parsing logic
- * No ownership drift
- * No unauthorized database ownership
- * No direct cross-service writes
- * No constitutional violations

Produce a Responsibility Matrix showing each service's constitutional role.

Phase 4 – End-to-End Runtime Certification

Validate the complete runtime:

Task Submission (PDF / DOCX / TXT / MD)

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NIYANTRAN

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PAIKSHAK

↓

MasterDB

↓

NIYANTRAN

Verify only that each handoff succeeds and contracts are honored. Do not re-test individual feature implementations owned by Ishan or Pritesh.

Phase 5 – Runtime Observability & Replay

Validate:

- * Trace IDs
- * Runtime logging
- * Replay support
- * Observability
- * Failure visibility
- * Audit trail

Every execution must be fully traceable.

Phase 6 – Production Certification

Issue a final Runtime Convergence Certificate confirming:

- * Runtime contracts verified
- * Service boundaries respected
- * Constitutional ownership maintained
- * No architectural drift
- * Production deployment approved

Final Deliverable

The primary artifact should be a TANTRA Runtime Convergence Certification Report, containing:

- * Runtime Dependency Matrix
- * Service Responsibility Matrix
- * Runtime Contract Validation
- * Architecture Drift Analysis
- * Integration Gap Report
- * Production Readiness Certification
- * Deployment Recommendation to Alay

This keeps the ownership perfectly clean:

- * Ishan builds the review runtime.
- * Pritesh builds the NIYANTRAN runtime and PRANA/KARMA integration.
- * KAVYA owns MasterDB.
- * Nupur owns MDU.
- * Rudra certifies that all of them function together as a single TANTRA runtime without taking ownership of their implementations.
- * Alay deploys the certified converged runtime to production.